United International University
Department of Computer Science and Engineering

CSE-3313: Computer Architecture
Final Examination: Summer 2025
Total Marks: 40 Time: 2 hours

Any examinee found adopting unfair means will be expelled from the trimester / program as per UIU disciplinary rules.

Answer all questions. Numbers to the right of the questions denote their marks.

1. (a) Modify and annotate the block diagram for the single-cycle data path and **write the control unit values** for this instruction: **'cat $s0, $s1, 1024'.** The task of this instruction is to read the value of $s1 **(found at register label rs)** and multiply it by 8 to get the first operand of the Arithmatic Logic Unit. The second operand of the Arithmatic Logic Unit will be 1024, which is stored at the offset. The ALU will calculate the sum of the two operands, and store the result at the address found at $s0 **(found at register label rt)** . The machine code format for cat is given below: [5]

op	rs	rt	offset
6 bits	5 bits	5 bits	16 bits

Table 1: Machine code format of instruction cat

(b) Modify and annotate the block diagram for a single-cycle data path so that it can execute the following custom R type instruction: **'cp $s0, $s1, 5'.** The task of this instruction is to read the value of $s1 and left shift it by 5. After that, the shift-resulted value will be compared with the value found at the address found at $s0. If the shift-resulted value is greater than the value found at the address, the program counter will jump next to an address equal to the difference between the two values compared. Assuming the opcode for cp is 0 and function code is 12 and the register number for $s0 and $s1 is 16 and 17, the machine code for this instruction, based on the instruction format given, will be given as table 2. **Write down the values for the control unit as well.** [5]

opcode (6 bits)	rs (5 bits)	rt (5 bits)	rd (5 bits)	shamt (5 bits)	function code (6 bits)
0	16	17		5	12

Table 2: Machine code translation of instruction cp $s0, $s1, 5

2. Consider a processor that goes through the following six stages while executing an instruction. Assume that when one instruction is using the memory stage to read or write, no other instruction can use the memory for write or another read in the same cycle.

Instruction Fetch (IF)	120 ns
Instruction Decode (ID)	150 ns
Register Read (RR)	150 ns
Execution (EXE)	180 ns
Memory	190 ns
Write Back (WB)	170 ns

Table 3: Six stages pipeline

a) Calculate the total execution time for the following code using the basic pipelining method. Provide a timing diagram. [4]

```
lw $t2, 32($s0)
and $t3, $t2, $t1
sw $t3, 36($s0)
lw $t4, 40($s0)
add $t4, $t2, $t3
```

b) Now for the code snippet given in (a), resolve all data hazards using the pipeline's hazard-handling policy and compute the total execution time. Provide a timing diagram. [4]

c) Consider the following MIPS assembly code segment. Assume that the base address of the array A is stored in \$s0, and A = [1, 2, -1, 4, 5, -6, 7, 8]. [4+3]

```
    addi $s1, $zero, 6
    addi $s2, $zero, 0
    addi $s3, $zero, 0
LOOP:
    beq  $s2, $s1, END
    sll  $t0, $s2, 2
    add  $t0, $t0, $s0
    lw   $t1, 0($t0)
    slt  $t2, $t1, $zero
    bne  $t2, $zero, SKIP
    add  $s3, $s3, $t1
SKIP:
    addi $s2, $s2, 1
    j    LOOP
END:
    sw   $s3, 0($t4)
```

i) Calculate the total execution time of the code using single-cycle processor.

ii) Determine the total number of pipeline stalls when the same code is executed on the given 6-stage pipelined processor with forwarding. Assume a 1-cycle stall for each taken branch or jump instruction.

3. (a) A CPU uses 2 kB direct-mapped cache with 8 words per block. The CPU generates 64-bit addresses and the memory is byte-addressable. Here, 1 word refers to 2 bytes.

i. Calculate the actual size of the cache in bits. [4]

ii. Find the miss ratio if the following memory addresses are accessed in this order. [8]
Assume the cache is initially empty.

80	96	120	88	16	99

(b) Does increasing block size always lead to a lower miss rate? Explain briefly with an example. [3]